Lecture Scribe CSE400

L9_S2_A.pdf
PDF

Generate a lecture scribe intended to serve as faithful, exam-ready reference material for revision.

Use only the provided context consisting of the lecture slides and relevant textbook excerpts. Do not introduce any material that does not appear in the given context.

While writing the scribe, reason step by step through all definitions, notation, assumptions, theorems or results, proofs or derivations, and worked example solutions exactly as presented in the context.

Make the logical flow between concepts explicit so that a student can clearly follow the reasoning during exam preparation.

Maintain a formal academic tone and structure the scribe clearly for revision purposes.

Below is a faithful, exam-ready lecture scribe derived **only** from the provided lecture slides for **CSE400 – Lecture 9 (Part shown)** on Uniform and Exponential Random Variables. All definitions, formulas, reasoning steps, and example solutions strictly follow the slide content as presented in the document .

CSE400 – Fundamentals of Probability in Computing

Lecture 9: Uniform and Exponential Random Variables

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1. Types of Continuous Random Variables (Overview)

The lecture introduces the following continuous random variables:

- Uniform Random Variable
- Exponential Random Variable

- Laplace Random Variable (listed but not developed in shown portion)
- Gamma Random Variable (listed but not developed in shown portion)

The focus in this material is on:

- Uniform Random Variable
- Exponential Random Variable

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2. Uniform Random Variable

Let X be a continuous random variable uniformly distributed over the interval $[a, b]$.

2.1 Probability Density Function (PDF)

$$f_X(x) = \begin{cases} \dfrac{1}{b-a}, & a \leqslant x < b \\ 0, & \text{elsewhere} \end{cases}$$

This means:

- The density is constant between a and b
- Zero outside the interval

2.2 Cumulative Distribution Function (CDF)

$$F_X(x) = \begin{cases} 0, & x < a \\ \dfrac{x-a}{b-a}, & a \leqslant x < b \\ 1, & x \geqslant b \end{cases}$$

Thus:

- Before a, probability is zero
- Between a and b, probability increases linearly
- After b, probability equals one

2.3 General Probability for Uniform RV on $[0, 2\pi]$

For a uniform random variable $\Theta \in [0, 2\pi]$:

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$$\Pr(a < \Theta < b) = \frac{b-a}{2\pi}$$

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3. Example 1 — Uniform Random Variable

Given:

The phase Θ of a sinusoid is uniformly distributed over $[0, 2\pi]$:

$$f_\Theta(\theta) = \begin{cases} \frac{1}{2\pi}, & 0 \leqslant \theta < 2\pi \\ 0, & \text{otherwise} \end{cases}$$

(a) Find $\Pr(\Theta > \frac{3\pi}{4})$

Using uniform probability:

$$\Pr(\Theta > \frac{3\pi}{4}) = \frac{2\pi - \frac{3\pi}{4}}{2\pi}$$
$$= \frac{8\pi - 3\pi}{8\pi} = \frac{5\pi}{8\pi} = \frac{5}{8}$$

(b) Find $\Pr(\Theta < \pi \mid \Theta > \frac{3\pi}{4})$

Use conditional probability:

$$\Pr(A \mid B) = \frac{\Pr(A \cap B)}{\Pr(B)}$$

Here:

- $B = \Theta > \frac{3\pi}{4}$
- $A = \Theta < \pi$

First compute:

$$\Pr\left(\frac{3\pi}{4} < \Theta < \pi\right) = \frac{\pi - \frac{3\pi}{4}}{2\pi} = \frac{\pi/4}{2\pi} = \frac{1}{8}$$

From part (a):

$$\Pr(B) = \frac{5}{8}$$

Therefore:

$$\Pr(\Theta < \pi \mid \Theta > \frac{3\pi}{4}) = \frac{1/8}{5/8} = \frac{1}{5}$$

(c) Find $\Pr(\cos\Theta < \frac{1}{2})$

Solve:

$$\cos\Theta = \frac{1}{2} \Rightarrow \Theta = \frac{\pi}{3}, \frac{5\pi}{3}$$

Thus:

$$\cos\Theta < \frac{1}{2} \quad \text{for} \quad \frac{\pi}{3} < \Theta < \frac{5\pi}{3}$$

Probability:

$$\Pr = \frac{\frac{5\pi}{3} - \frac{\pi}{3}}{2\pi} = \frac{4\pi/3}{2\pi} = \frac{2}{3}$$

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4. Applications of Uniform Random Variable

Examples stated:

- Phase of a sinusoidal signal uniformly between 0 and 2π
- Random number between 0 and 1 in simulations
- Arrival time within a known time window assuming no preference

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5. Exponential Random Variable

Let $b > 0$.

5.1 Probability Density Function (PDF)

$$f_X(x) = \frac{1}{b} \exp\left(-\frac{x}{b}\right) u(x)$$

where $u(x)$ is the unit step function ensuring:

$$x \geqslant 0$$

5.2 Cumulative Distribution Function (CDF)

$$F_X(x) = [1 - \exp(-\frac{x}{b})]\, u(x)$$

Thus:

- Probability starts at zero
- Increases exponentially toward 1 as $x \to \infty$

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6. Example 2 — Exponential Random Variable

Given:

$$f_X(x) = e^{-x} u(x)$$

This corresponds to exponential RV with $b = 1$.

(a) Find $\Pr(3X < 5)$

First rewrite inequality:

$$3X < 5 \Rightarrow X < \frac{5}{3}$$

Now use CDF:

$$\Pr(X < y) = 1 - e^{-y}$$

So:

$$\Pr(3X < 5) = 1 - e^{-5/3}$$

(b) Generalize: Find $\Pr(3X < y)$

Rewrite:

$$3X < y \Rightarrow X < \frac{y}{3}$$

Thus:

$$\Pr(3X < y) = 1 - e^{-y/3}$$

End of Provided Lecture Content

(Only Uniform and Exponential RVs were developed in the supplied material.)

If you'd like, I can:

✅ Convert this scribe into LaTeX exam notes
✅ Make a formula sheet
✅ Add practice problems (only from this lecture's style)
✅ Continue with Laplace & Gamma when you upload those slides

Just tell me 👍